

Table S1: Modification detection tools benchmarked in this study.

Tool	Classification strategy	Classification method	Specific algorithms	Targeted modification	Modes evaluated in this study
m6Anet	Signal (current)	Deep learning	Multiple-instance-learning neural network	m6A	de novo (RNA002 and RNA004-optimised models)
MINES	Signal (current)	Machine learning	Random forest	m6A	de novo
DENA	Signal (current)	Deep learning	Bidirectional LSTM	m6A	de novo
Nanom6A	Signal (current)	Machine learning	XGBoost	m6A	de novo
Yanocomp	Signal (current)	Machine learning + statistical test	Gaussian mixture model, G-test	m6A	de novo
xPore	Signal (current)	Clustering	Multi-sample two-Gaussian mixture model	m6A	comparative (WT vs control)
DRUMMER	Basecalling errors	Statistical test	Comparative profiling of basecall error rates, G-test	m6A	comparative (WT vs control)
ELIGOS2	Basecalling errors	Statistical test	Fisher's exact test	m6A, m5C	differential (ELIGOS2_diff) and single-sample (ELIGOS2_solo)
EpiNano	Basecalling errors	Machine learning	Support vector machine (error features)	m6A	error-based mode (EpiNano_Error)
Nanocompore	Signal (current)	Machine learning + statistical test	Two-GMM followed by logit test	m6A	comparative (WT vs control)
CHEUI	Signal (current)	Deep learning + statistical test	Convolutional neural network, two-tailed Mann-Whitney U-test	m6A, m5C	CHEUI_m6A, CHEUI_m5C
NanoSPA	Signal (current)	Machine learning	Feedforward neural network	m6A, Psi	NanoSPA_m6A, NanoSPA_Psi
NanoMUD	Signal (current)	Machine learning + deep learning	Bidirectional LSTM, regression and motif-specific models	Psi, m1Psi	NanoMUD_Psi, NanoMUD_m1Psi
NanoPsu	Basecalling errors	Machine learning + statistical test	U-to-C mismatch error analysis, EXT	Psi	de novo
NanoNm	Signal (current)	Machine learning	XGBoost	Nm	de novo

Note. RNA002 datasets for all tools; Dorado built-in modification models (RNA004) and the RNA004-optimised m6Anet model are reported separately in Fig. 8 / Fig. S8. Tool names follow the standardized nomenclature of the manuscript; versions and command lines are in Table S6.

Table S2: m6A sites detected in the three independent HeLa sequencing units per tool.

Condition	Tool	Unit 1	Unit 2	Unit 3	Mean	SD	CV
HeLa WT	ELIGOS2_solo	49814	58726	35520	48020.0	11707.0	0.244
HeLa WT	MINES	31998	61460	26272	39910.0	18881.0	0.473
HeLa WT	EpiNano_Error	24182	46813	18904	29966.0	14826.0	0.495
HeLa WT	xPore	23881	39851	19952	27895.0	10539.0	0.378
HeLa WT	CHEUI_m6A	19932	31096	16812	22613.0	7510.0	0.332
HeLa WT	Nanom6A	20892	18286	17048	18742.0	1962.0	0.105
HeLa WT	DENA	17473	17352	14665	16497.0	1587.0	0.096
HeLa WT	m6Anet	7437	13883	5611	8977.0	4346.0	0.484
HeLa WT	yanocomp	4614	6842	4338	5265.0	1373.0	0.261
HeLa WT	NanoSPA_m6A	1375	2490	941	1602.0	799.0	0.499
HeLa WT	Nanocompore	1079	1534	1181	1265.0	239.0	0.189
HeLa WT	ELIGOS2_diff	1065	1816	820	1234.0	519.0	0.421
HeLa WT	DRUMMER	367	637	255	420.0	196.0	0.468
HeLa IVT	MINES	42134	64266	30215	45538.0	17279.0	0.379
HeLa IVT	ELIGOS2_solo	45923	58353	22995	42424.0	17937.0	0.423
HeLa IVT	xPore	34645	25623	27252	29173.0	4808.0	0.165
HeLa IVT	EpiNano_Error	33972	29312	20905	28063.0	6622.0	0.236
HeLa IVT	CHEUI_m6A	23527	37544	13473	24848.0	12090.0	0.487
HeLa IVT	DENA	16121	22584	9009	15905.0	6790.0	0.427
HeLa IVT	Nanom6A	12644	15078	11431	13051.0	1857.0	0.142
HeLa IVT	m6Anet	5996	11536	4528	7353.0	3696.0	0.503
HeLa IVT	yanocomp	3569	1867	3504	2980.0	964.0	0.324
HeLa IVT	NanoSPA_m6A	408	227	230	288.0	104.0	0.359
HeLa IVT	Nanocompore	105	59	201	122.0	72.0	0.595
HeLa IVT	ELIGOS2_diff	27	82	72	60.0	29.0	0.486
HeLa IVT	DRUMMER	19	8	69	32.0	33.0	1.016

Note. One row per tool and condition; each unit is an independent sequencing unit (HeLa_WT1-3, HeLa_IVT_rep1-3). The values are those plotted in Fig. 1B and are deposited in the machine-readable summary table that accompanies the source code.

Table S3: Wild-type versus modification-deficient calls on the shared testable universe.

Species	Tool	Pair	Shared universe (positions)	WT calls	Deficient-condition calls	Deficient/WT ratio
Arabidopsis	CHEUI_m6A	rep1	3346442	13355	14372	1.076
Arabidopsis	CHEUI_m6A	rep2	4364433	18390	11450	0.623
Arabidopsis	CHEUI_m6A	rep3	2478089	12510	10218	0.817
Arabidopsis	CHEUI_m6A	mean				0.839
Arabidopsis	DENA	rep1	3346442	16645	9264	0.557
Arabidopsis	DENA	rep2	4364433	16552	13054	0.789
Arabidopsis	DENA	rep3	2478089	12769	5554	0.435
Arabidopsis	DENA	mean				0.593
Arabidopsis	DRUMMER	rep1	3346442	463	16	0.035
Arabidopsis	DRUMMER	rep2	4364433	597	0	0.0
Arabidopsis	DRUMMER	rep3	2478089	324	19	0.059
Arabidopsis	DRUMMER	mean				0.031
Arabidopsis	ELIGOS2_diff	rep1	3346442	1003	13	0.013
Arabidopsis	ELIGOS2_diff	rep2	4364433	1243	24	0.019
Arabidopsis	ELIGOS2_diff	rep3	2478089	719	7	0.01
Arabidopsis	ELIGOS2_diff	mean				0.014
Arabidopsis	ELIGOS2_solo	rep1	3346442	35781	20745	0.58
Arabidopsis	ELIGOS2_solo	rep2	4364433	33393	26729	0.8
Arabidopsis	ELIGOS2_solo	rep3	2478089	26321	10245	0.389
Arabidopsis	ELIGOS2_solo	mean				0.59
Arabidopsis	EpiNano_Error	rep1	3346442	9235	8285	0.897
Arabidopsis	EpiNano_Error	rep2	4364433	14251	4993	0.35
Arabidopsis	EpiNano_Error	rep3	2478089	5488	4741	0.864
Arabidopsis	EpiNano_Error	mean				0.704
Arabidopsis	MINES	rep1	3346442	50099	27838	0.556
Arabidopsis	MINES	rep2	4364433	47729	40888	0.857
Arabidopsis	MINES	rep3	2478089	40449	19006	0.47
Arabidopsis	MINES	mean				0.627
Arabidopsis	NanoSPA_m6A	rep1	3346442	745	245	0.329
Arabidopsis	NanoSPA_m6A	rep2	4364433	950	391	0.412
Arabidopsis	NanoSPA_m6A	rep3	2478089	644	182	0.283
Arabidopsis	NanoSPA_m6A	mean				0.341
Arabidopsis	Nanocompore	rep1	3346442	443	191	0.431
Arabidopsis	Nanocompore	rep2	4364433	1070	160	0.15
Arabidopsis	Nanocompore	rep3	2478089	245	661	2.698

Species	Tool	Pair	Shared universe (positions)	WT calls	Deficient-condition calls	Deficient/WT ratio
Arabidopsis	Nanocompore	mean				1.093
Arabidopsis	Nanom6A	rep1	3346442	20630	15842	0.768
Arabidopsis	Nanom6A	rep2	4364433	19673	15015	0.763
Arabidopsis	Nanom6A	rep3	2478089	17309	12530	0.724
Arabidopsis	Nanom6A	mean				0.752
Arabidopsis	m6Anet	rep1	3346442	9364	2566	0.274
Arabidopsis	m6Anet	rep2	4364433	10015	4268	0.426
Arabidopsis	m6Anet	rep3	2478089	6712	1688	0.251
Arabidopsis	m6Anet	mean				0.317
Arabidopsis	xPore	rep1	3346442	27049	20949	0.774
Arabidopsis	xPore	rep2	4364433	37680	22000	0.584
Arabidopsis	xPore	rep3	2478089	20904	22544	1.078
Arabidopsis	xPore	mean				0.812
Arabidopsis	yanocomp	rep1	3346442	3578	4167	1.165
Arabidopsis	yanocomp	rep2	4364433	13031	2081	0.16
Arabidopsis	yanocomp	rep3	2478089	2058	5011	2.435
Arabidopsis	yanocomp	mean				1.253
Human	CHEUI_m6A	rep1	2767950	16871	20153	1.195
Human	CHEUI_m6A	rep2	4308139	25484	33110	1.299
Human	CHEUI_m6A	rep3	1959262	14499	11366	0.784
Human	CHEUI_m6A	mean				1.093
Human	DENA	rep1	2767950	14153	13213	0.934
Human	DENA	rep2	4308139	12819	19219	1.499
Human	DENA	rep3	1959262	11250	7304	0.649
Human	DENA	mean				1.027
Human	DRUMMER	rep1	2767950	358	19	0.053
Human	DRUMMER	rep2	4308139	607	8	0.013
Human	DRUMMER	rep3	1959262	249	69	0.277
Human	DRUMMER	mean				0.114
Human	ELIGOS2_diff	rep1	2767950	958	9	0.009
Human	ELIGOS2_diff	rep2	4308139	1634	64	0.039
Human	ELIGOS2_diff	rep3	1959262	740	43	0.058
Human	ELIGOS2_diff	mean				0.036
Human	ELIGOS2_solo	rep1	2767950	34095	32290	0.947
Human	ELIGOS2_solo	rep2	4308139	38492	44129	1.146
Human	ELIGOS2_solo	rep3	1959262	23148	16129	0.697
Human	ELIGOS2_solo	mean				0.93

Species	Tool	Pair	Shared universe (positions)	WT calls	Deficient-condition calls	Deficient/WT ratio
Human	EpiNano_Error	rep1	2767950	19451	19831	1.02
Human	EpiNano_Error	rep2	4308139	38137	23961	0.628
Human	EpiNano_Error	rep3	1959262	15774	12681	0.804
Human	EpiNano_Error	mean				0.817
Human	MINES	rep1	2767950	26868	34757	1.294
Human	MINES	rep2	4308139	47977	54981	1.146
Human	MINES	rep3	1959262	21322	24483	1.148
Human	MINES	mean				1.196
Human	NanoSPA_m6A	rep1	2767950	769	219	0.285
Human	NanoSPA_m6A	rep2	4308139	1217	117	0.096
Human	NanoSPA_m6A	rep3	1959262	446	111	0.249
Human	NanoSPA_m6A	mean				0.21
Human	Nanocompore	rep1	2767950	1051	97	0.092
Human	Nanocompore	rep2	4308139	1471	59	0.04
Human	Nanocompore	rep3	1959262	1164	195	0.168
Human	Nanocompore	mean				0.1
Human	Nanom6A	rep1	2767950	17529	11612	0.662
Human	Nanom6A	rep2	4308139	15060	14364	0.954
Human	Nanom6A	rep3	1959262	13742	9900	0.72
Human	Nanom6A	mean				0.779
Human	m6Anet	rep1	2767950	6150	5059	0.823
Human	m6Anet	rep2	4308139	10768	9979	0.927
Human	m6Anet	rep3	1959262	4485	3790	0.845
Human	m6Anet	mean				0.865
Human	xPore	rep1	2767950	23313	30213	1.296
Human	xPore	rep2	4308139	38853	24299	0.625
Human	xPore	rep3	1959262	19559	22681	1.16
Human	xPore	mean				1.027
Human	yanocomp	rep1	2767950	4449	3206	0.721
Human	yanocomp	rep2	4308139	6547	1770	0.27
Human	yanocomp	rep3	1959262	4240	3010	0.71
Human	yanocomp	mean				0.567
Mouse	CHEUI_m6A	SRP166020	2586167	10238	15957	1.559
Mouse	CHEUI_m6A	SRP357195	6666647	36581	32217	0.881
Mouse	CHEUI_m6A	mean				1.22
Mouse	DENA	SRP166020	2586167	22780	10979	0.482
Mouse	DENA	SRP357195	6666647	34823	22632	0.65

Species	Tool	Pair	Shared universe (positions)	WT calls	Deficient-condition calls	Deficient/WT ratio
Mouse	DENA	mean				0.566
Mouse	DRUMMER	SRP166020	2586167	510	0	0.0
Mouse	DRUMMER	SRP357195	6666647	1831	407	0.222
Mouse	DRUMMER	mean				0.111
Mouse	ELIGOS2_diff	SRP166020	2586167	817	0	0.0
Mouse	ELIGOS2_diff	SRP357195	6666647	3865	580	0.15
Mouse	ELIGOS2_diff	mean				0.075
Mouse	ELIGOS2_solo	SRP166020	2586167	50915	24874	0.489
Mouse	ELIGOS2_solo	SRP357195	6666647	77063	50674	0.658
Mouse	ELIGOS2_solo	mean				0.573
Mouse	EpiNano_Error	SRP166020	2586167	15903	18129	1.14
Mouse	EpiNano_Error	SRP357195	6666647	25833	0	0.0
Mouse	EpiNano_Error	mean				0.57
Mouse	MINES	SRP166020	2586167	49607	29280	0.59
Mouse	MINES	SRP357195	6666647	82499	85278	1.034
Mouse	MINES	mean				0.812
Mouse	NanoSPA_m6A	SRP166020	2586167	2405	551	0.229
Mouse	NanoSPA_m6A	SRP357195	6666647	2706	127	0.047
Mouse	NanoSPA_m6A	mean				0.138
Mouse	Nanocompore	SRP166020	2586167	918	7217	7.862
Mouse	Nanocompore	SRP357195	6666647	3022	0	0.0
Mouse	Nanocompore	mean				3.931
Mouse	Nanom6A	SRP166020	2586167	10213	7686	0.753
Mouse	Nanom6A	SRP357195	6666647	23746	18767	0.79
Mouse	Nanom6A	mean				0.771
Mouse	m6Anet	SRP166020	2586167	18639	6313	0.339
Mouse	m6Anet	SRP357195	6666647	26218	15305	0.584
Mouse	m6Anet	mean				0.461
Mouse	xPore	SRP166020	2586167	40639	0	0.0
Mouse	xPore	SRP357195	6666647	59195	43192	0.73
Mouse	xPore	mean				0.365
Mouse	yanocomp	SRP166020	2586167	8723	0	0.0
Mouse	yanocomp	SRP357195	6666647	7216	43985	6.095
Mouse	yanocomp	mean				3.048

Note. The shared testable universe of a matched pair is the set of exonic positions with coverage ≥ 10 and a modification-compatible reference base in BOTH samples; the ratio is the deficient-condition call count over the wild-type call count on that universe (Fig. 2A). Mouse pairs are one per independent study and are never pooled. Deficient samples are fip37 KD (Arabidopsis), Mettl3 KO (mouse) and the unmodified IVT library (HeLa); KO/KD are partial negatives, so the ratio is a relative enrichment change, not a specificity measure. The per-pair values are deposited in the machine-readable table that accompanies the source code.

Table S4: Five most frequently detected 5-mer motifs per tool and species.

Species	Tool	Rank	5-mer	Count	Relative frequency	RRACH
Arabidopsis	CHEUI_m6A	1	GAAGA	842	0.0151	no
Arabidopsis	CHEUI_m6A	2	AGAAG	694	0.0124	no
Arabidopsis	CHEUI_m6A	3	AGAGA	593	0.0106	no
Arabidopsis	CHEUI_m6A	4	UGAAG	578	0.0103	no
Arabidopsis	CHEUI_m6A	5	AAAGA	557	0.01	no
Arabidopsis	DENA	1	AAACA	17689	0.2647	yes
Arabidopsis	DENA	2	AAACU	11552	0.1733	yes
Arabidopsis	DENA	3	GAACU	8931	0.1341	yes
Arabidopsis	DENA	4	GAACA	6604	0.0986	yes
Arabidopsis	DENA	5	GAACC	6370	0.0956	yes
Arabidopsis	DRUMMER	1	UAAGA	83	0.0576	no
Arabidopsis	DRUMMER	2	UCAGA	72	0.049	no
Arabidopsis	DRUMMER	3	UGACA	64	0.0447	no
Arabidopsis	DRUMMER	4	UUAGA	63	0.044	no
Arabidopsis	DRUMMER	5	UGAGA	63	0.0438	no
Arabidopsis	ELIGOS_diff	1	UAAGA	129	0.0431	no
Arabidopsis	ELIGOS_diff	2	AGACA	132	0.043	yes
Arabidopsis	ELIGOS_diff	3	UCAGA	125	0.0419	no
Arabidopsis	ELIGOS_diff	4	AAACA	91	0.0304	yes
Arabidopsis	ELIGOS_diff	5	UUAGA	84	0.0281	no
Arabidopsis	ELIGOS_solo	1	UCACC	6402	0.0434	no
Arabidopsis	ELIGOS_solo	2	UCACU	6010	0.0408	no
Arabidopsis	ELIGOS_solo	3	ACAUU	4166	0.0279	no
Arabidopsis	ELIGOS_solo	4	UCACA	4133	0.0278	no
Arabidopsis	ELIGOS_solo	5	CCACC	3847	0.026	no
Arabidopsis	EpiNano_Error	1	UCAA	599	0.02	no
Arabidopsis	EpiNano_Error	2	UUAAA	479	0.0163	no
Arabidopsis	EpiNano_Error	3	ACAAA	482	0.0161	no
Arabidopsis	EpiNano_Error	4	AUAAA	489	0.0158	no
Arabidopsis	EpiNano_Error	5	AGAAA	481	0.0158	no
Arabidopsis	MINES	1	AAACA	25982	0.1364	yes
Arabidopsis	MINES	2	AAACC	23405	0.1229	yes
Arabidopsis	MINES	3	AGACA	21556	0.1134	yes
Arabidopsis	MINES	4	AAACU	21500	0.1131	yes
Arabidopsis	MINES	5	GAACA	17927	0.0943	yes

Species	Tool	Rank	5-mer	Count	Relative frequency	RRACH
Arabidopsis	NanoSPA_m6A	1	AGACU	1097	0.2649	yes
Arabidopsis	NanoSPA_m6A	2	AGACA	619	0.1502	yes
Arabidopsis	NanoSPA_m6A	3	UGACU	571	0.1391	no
Arabidopsis	NanoSPA_m6A	4	GGACU	479	0.1162	yes
Arabidopsis	NanoSPA_m6A	5	GGACA	229	0.0552	yes
Arabidopsis	Nanocompore	1	AGACU	260	0.1854	yes
Arabidopsis	Nanocompore	2	AGACA	198	0.1398	yes
Arabidopsis	Nanocompore	3	AGACC	51	0.034	yes
Arabidopsis	Nanocompore	4	AAACU	56	0.0334	yes
Arabidopsis	Nanocompore	5	AGACG	38	0.0283	no
Arabidopsis	Nanom6A	1	AAACA	7988	0.1307	yes
Arabidopsis	Nanom6A	2	AAACU	7758	0.1268	yes
Arabidopsis	Nanom6A	3	AAACC	6634	0.1078	yes
Arabidopsis	Nanom6A	4	AGACA	4853	0.0784	yes
Arabidopsis	Nanom6A	5	GGACA	3887	0.0632	yes
Arabidopsis	m6Anet	1	GAACU	9785	0.2743	yes
Arabidopsis	m6Anet	2	AAACU	4496	0.1264	yes
Arabidopsis	m6Anet	3	UGACU	4368	0.1227	no
Arabidopsis	m6Anet	4	GAACA	2956	0.0826	yes
Arabidopsis	m6Anet	5	GGACU	2808	0.0794	yes
Arabidopsis	xPore	1	GAAGA	1359	0.0159	no
Arabidopsis	xPore	2	ACAAG	1206	0.0133	no
Arabidopsis	xPore	3	AAACU	1006	0.0116	yes
Arabidopsis	xPore	4	AAAUG	968	0.0114	no
Arabidopsis	xPore	5	AGAAG	956	0.0112	no
Arabidopsis	Yanocomp	1	AAACU	425	0.0308	yes
Arabidopsis	Yanocomp	2	AAAAA	264	0.0171	no
Arabidopsis	Yanocomp	3	AAAAC	238	0.0166	no
Arabidopsis	Yanocomp	4	GGAUU	241	0.0159	no
Arabidopsis	Yanocomp	5	AGACU	247	0.0157	yes
E.coli	CHEUI_m6A	1	UGAUG	230	0.0121	no
E.coli	CHEUI_m6A	2	UGAAA	227	0.012	no
E.coli	CHEUI_m6A	3	GAAAA	210	0.0112	no
E.coli	CHEUI_m6A	4	GAAGA	205	0.0106	no
E.coli	CHEUI_m6A	5	UGAAG	199	0.0101	no
E.coli	DRUMMER	1	CGAAA	16	0.1716	no
E.coli	DRUMMER	2	GGAAC	14	0.1501	no

Species	Tool	Rank	5-mer	Count	Relative frequency	RRACH
E.coli	DRUMMER	3	CGACC	10	0.1181	no
E.coli	DRUMMER	4	ACACC	6	0.0535	no
E.coli	DRUMMER	5	AUAAG	7	0.0372	no
E.coli	ELIGOS_diff	1	GAACC	30	0.0833	yes
E.coli	ELIGOS_diff	2	AUAAC	27	0.0751	no
E.coli	ELIGOS_diff	3	CGAUG	15	0.0417	no
E.coli	ELIGOS_diff	4	AGACA	14	0.0389	yes
E.coli	ELIGOS_diff	5	CUAUA	14	0.0389	no
E.coli	ELIGOS_solo	1	UCACC	1438	0.0453	no
E.coli	ELIGOS_solo	2	CAACC	854	0.0269	no
E.coli	ELIGOS_solo	3	UCACU	839	0.0264	no
E.coli	ELIGOS_solo	4	CAACU	679	0.0214	no
E.coli	ELIGOS_solo	5	ACAUU	666	0.021	no
E.coli	EpiNano_Error	1	UGAAA	257	0.0356	no
E.coli	EpiNano_Error	2	CGAAA	232	0.0332	no
E.coli	EpiNano_Error	3	GCAAA	185	0.0265	no
E.coli	EpiNano_Error	4	UCACC	164	0.0235	no
E.coli	EpiNano_Error	5	GUAAA	135	0.019	no
E.coli	NanoSPA_m6A	1	UGACU	110	0.2256	no
E.coli	NanoSPA_m6A	2	AGACU	93	0.1974	yes
E.coli	NanoSPA_m6A	3	GGACU	72	0.1542	yes
E.coli	NanoSPA_m6A	4	GGACG	47	0.102	no
E.coli	NanoSPA_m6A	5	GGACC	15	0.0329	yes
E.coli	Nanocompore	1	GUACU	104	0.0255	no
E.coli	Nanocompore	2	GCAAC	87	0.0213	no
E.coli	Nanocompore	3	CAAAA	72	0.0176	no
E.coli	Nanocompore	4	ACAUC	69	0.0169	no
E.coli	Nanocompore	5	ACAAC	66	0.0162	no
E.coli	Nanom6A	1	AAACG	2108	0.1376	no
E.coli	Nanom6A	2	AAACU	1569	0.1035	yes
E.coli	Nanom6A	3	AAACC	1584	0.1032	yes
E.coli	Nanom6A	4	GAACG	1395	0.0915	no
E.coli	Nanom6A	5	AAACA	1192	0.0775	yes
E.coli	m6Anet	1	GAACU	2993	0.5509	yes
E.coli	m6Anet	2	GAACA	778	0.1433	yes
E.coli	m6Anet	3	UGACU	775	0.1432	no
E.coli	m6Anet	4	GAACC	295	0.0547	yes

Species	Tool	Rank	5-mer	Count	Relative frequency	RRACH
E.coli	m6Anet	5	GGACU	240	0.0439	yes
E.coli	xPore	1	UAAAG	215	0.0172	no
E.coli	xPore	2	CAAAA	195	0.0154	no
E.coli	xPore	3	CCAGC	180	0.0144	no
E.coli	xPore	4	UAAAA	170	0.0136	no
E.coli	xPore	5	CAAAG	156	0.0123	no
E.coli	Yanocomp	1	GCAAC	145	0.0162	no
E.coli	Yanocomp	2	CGAUG	109	0.0132	no
E.coli	Yanocomp	3	UGAUG	107	0.0124	no
E.coli	Yanocomp	4	UUAUC	107	0.0124	no
E.coli	Yanocomp	5	GUACC	105	0.0119	no
Human	CHEUI_m6A	1	AAAAA	849	0.0122	no
Human	CHEUI_m6A	2	GAAGA	734	0.0108	no
Human	CHEUI_m6A	3	AGAAG	704	0.0103	no
Human	CHEUI_m6A	4	GGAGG	669	0.0098	no
Human	CHEUI_m6A	5	UGAAG	642	0.0095	no
Human	DENA	1	AAACA	13139	0.2239	yes
Human	DENA	2	AAACU	7776	0.1327	yes
Human	DENA	3	GAACA	5690	0.0974	yes
Human	DENA	4	GAACC	5630	0.0965	yes
Human	DENA	5	GAACU	4169	0.0741	yes
Human	DRUMMER	1	GGACU	280	0.1802	yes
Human	DRUMMER	2	GGACA	210	0.1321	yes
Human	DRUMMER	3	AAAGA	131	0.0923	no
Human	DRUMMER	4	AGACU	102	0.0691	yes
Human	DRUMMER	5	GGACC	104	0.0682	yes
Human	ELIGOS_diff	1	GGACU	606	0.1608	yes
Human	ELIGOS_diff	2	AAAAA	494	0.1354	no
Human	ELIGOS_diff	3	GGACA	413	0.1098	yes
Human	ELIGOS_diff	4	GGACC	239	0.0627	yes
Human	ELIGOS_diff	5	AGACU	180	0.0466	yes
Human	ELIGOS_solo	1	UCACC	6667	0.0463	no
Human	ELIGOS_solo	2	CCACC	5346	0.0376	no
Human	ELIGOS_solo	3	UCACU	5015	0.0343	no
Human	ELIGOS_solo	4	AAAAA	3820	0.0251	no
Human	ELIGOS_solo	5	UCACA	3379	0.0233	no
Human	EpiNano_Error	1	AAAAA	1584	0.0172	no

Species	Tool	Rank	5-mer	Count	Relative frequency	RRACH
Human	EpiNano_Error	2	UUAAA	1208	0.0135	no
Human	EpiNano_Error	3	CCACC	1153	0.0133	no
Human	EpiNano_Error	4	AGAAA	1099	0.0123	no
Human	EpiNano_Error	5	AUAAA	1046	0.012	no
Human	MINES	1	AAACA	16902	0.1114	yes
Human	MINES	2	GGACA	15663	0.1054	yes
Human	MINES	3	AGACA	14725	0.0972	yes
Human	MINES	4	AAACU	14103	0.0922	yes
Human	MINES	5	GGACC	13055	0.0886	yes
Human	NanoSPA_m6A	1	GGACU	1271	0.2765	yes
Human	NanoSPA_m6A	2	AGACU	1009	0.2007	yes
Human	NanoSPA_m6A	3	GGACA	483	0.0967	yes
Human	NanoSPA_m6A	4	UGACU	351	0.0748	no
Human	NanoSPA_m6A	5	GGACC	337	0.0716	yes
Human	Nanocompore	1	GGACU	590	0.1304	yes
Human	Nanocompore	2	AGACU	327	0.0733	yes
Human	Nanocompore	3	AUAAA	190	0.0435	no
Human	Nanocompore	4	GGACA	163	0.036	yes
Human	Nanocompore	5	AAAAA	93	0.0207	no
Human	Nanom6A	1	AAACA	8654	0.1533	yes
Human	Nanom6A	2	GGACU	7220	0.1286	yes
Human	Nanom6A	3	AAACU	6795	0.1208	yes
Human	Nanom6A	4	GGACA	6440	0.1145	yes
Human	Nanom6A	5	GAACU	4761	0.0846	yes
Human	m6Anet	1	GAACU	7538	0.2334	yes
Human	m6Anet	2	GGACU	6158	0.1869	yes
Human	m6Anet	3	UGACU	4549	0.1424	no
Human	m6Anet	4	GGACA	2810	0.0869	yes
Human	m6Anet	5	GAACA	2334	0.0728	yes
Human	xPore	1	GGACA	1418	0.0159	yes
Human	xPore	2	GGACU	1408	0.0155	yes
Human	xPore	3	AAAUG	1310	0.0146	no
Human	xPore	4	GAAGA	1117	0.0126	no
Human	xPore	5	AGAAG	1089	0.0123	no
Human	Yanocomp	1	GGACU	527	0.0321	yes
Human	Yanocomp	2	AAAAA	309	0.0189	no
Human	Yanocomp	3	GGACA	289	0.0174	yes

Species	Tool	Rank	5-mer	Count	Relative frequency	RRACH
Human	Yanocomp	4	GAAAA	207	0.013	no
Human	Yanocomp	5	GAACU	194	0.0118	yes
Mouse	CHEUI_m6A	1	GAAGA	475	0.0128	no
Mouse	CHEUI_m6A	2	AGAAG	457	0.0123	no
Mouse	CHEUI_m6A	3	GGAGG	391	0.0105	no
Mouse	CHEUI_m6A	4	AAAAA	387	0.0104	no
Mouse	CHEUI_m6A	5	UGAAG	376	0.0101	no
Mouse	DENA	1	AAACA	6864	0.1773	yes
Mouse	DENA	2	GAACU	5032	0.13	yes
Mouse	DENA	3	AAACU	3806	0.0983	yes
Mouse	DENA	4	GGACU	3665	0.0947	yes
Mouse	DENA	5	GGACA	3393	0.0876	yes
Mouse	DRUMMER	1	GGACU	428	0.2241	yes
Mouse	DRUMMER	2	GGACA	318	0.1665	yes
Mouse	DRUMMER	3	AGACU	154	0.0806	yes
Mouse	DRUMMER	4	GGACC	154	0.0806	yes
Mouse	DRUMMER	5	AAAGA	95	0.0497	no
Mouse	ELIGOS_diff	1	GGACU	847	0.2161	yes
Mouse	ELIGOS_diff	2	GGACA	482	0.123	yes
Mouse	ELIGOS_diff	3	GGACC	315	0.0804	yes
Mouse	ELIGOS_diff	4	AGACU	261	0.0666	yes
Mouse	ELIGOS_diff	5	UGACU	196	0.05	no
Mouse	ELIGOS_solo	1	UCACC	3660	0.042	no
Mouse	ELIGOS_solo	2	UCACU	3058	0.0351	no
Mouse	ELIGOS_solo	3	CCACC	2634	0.0302	no
Mouse	ELIGOS_solo	4	UCACA	2505	0.0287	no
Mouse	ELIGOS_solo	5	GGACU	2071	0.0237	yes
Mouse	EpiNano_Error	1	GGACU	529	0.016	yes
Mouse	EpiNano_Error	2	GGACA	464	0.014	yes
Mouse	EpiNano_Error	3	AGAAA	460	0.0139	no
Mouse	EpiNano_Error	4	CCACA	448	0.0135	no
Mouse	EpiNano_Error	5	UUAAA	443	0.0134	no
Mouse	MINES	1	AGACA	11325	0.1215	yes
Mouse	MINES	2	GGACA	10653	0.1143	yes
Mouse	MINES	3	AAACA	9534	0.1023	yes
Mouse	MINES	4	AGACC	8220	0.0882	yes
Mouse	MINES	5	AAACU	7469	0.0801	yes

Species	Tool	Rank	5-mer	Count	Relative frequency	RRACH
Mouse	NanoSPA_m6A	1	GGACU	1090	0.3336	yes
Mouse	NanoSPA_m6A	2	AGACU	764	0.2339	yes
Mouse	NanoSPA_m6A	3	GGACA	343	0.105	yes
Mouse	NanoSPA_m6A	4	UGACU	276	0.0845	no
Mouse	NanoSPA_m6A	5	GGACC	189	0.0579	yes
Mouse	Nanocompore	1	GGACU	849	0.2726	yes
Mouse	Nanocompore	2	AGACU	651	0.2091	yes
Mouse	Nanocompore	3	GGACA	283	0.0909	yes
Mouse	Nanocompore	4	GGACC	159	0.0511	yes
Mouse	Nanocompore	5	AGACA	124	0.0398	yes
Mouse	Nanom6A	1	GGACU	3418	0.1428	yes
Mouse	Nanom6A	2	AAACA	3305	0.138	yes
Mouse	Nanom6A	3	GGACA	2657	0.111	yes
Mouse	Nanom6A	4	AAACU	2471	0.1032	yes
Mouse	Nanom6A	5	AGACU	2188	0.0914	yes
Mouse	m6Anet	1	GAACU	6523	0.2232	yes
Mouse	m6Anet	2	GGACU	5328	0.1823	yes
Mouse	m6Anet	3	UGACU	4203	0.1438	no
Mouse	m6Anet	4	GGACA	2789	0.0954	yes
Mouse	m6Anet	5	AGACU	2315	0.0792	yes
Mouse	xPore	1	GGACU	1963	0.0327	yes
Mouse	xPore	2	GGACA	1403	0.0233	yes
Mouse	xPore	3	AGACU	1270	0.0211	yes
Mouse	xPore	4	GGACC	847	0.0141	yes
Mouse	xPore	5	GAAGA	713	0.0119	no
Mouse	Yanocomp	1	GGACU	706	0.0977	yes
Mouse	Yanocomp	2	GGACA	278	0.0385	yes
Mouse	Yanocomp	3	AGACU	254	0.0352	yes
Mouse	Yanocomp	4	GAACU	198	0.0274	yes
Mouse	Yanocomp	5	GGACC	149	0.0206	yes

Note. Motifs are counted on the strand-normalized 5-mers centred on each called A of the cleaned call set, with equal weight per independent unit; the relative frequency is the replicate-mixed frequency within the tool x species 5-mer pool (Fig. S2B). Mouse uses the mES_WT study only; E. coli has no DENA or MINES run. The full per-tool table is deposited with the source code.

Table S5: Non-m6A calls in HeLa and the unmodified-IVT-to-WT ratio.

Tool	Modification	WT union	Unmodified IVT union	IVT/WT ratio (unions)	Mean-of-counts ratio	95% CI low	95% CI high
CHEUI-m5C	m5C	47747	51171	1.0717	1.081	0.635	1.638
NanoMUD-Psi	Psi	13711	14794	1.079	1.155	0.725	1.87
NanoMUD-m1Psi	m1Psi	40656	43063	1.0592	1.164	0.688	2.031
NanoNm	Nm	4181	6399	1.5305	1.714	0.849	2.993
NanoSPA-Psi	Psi	888	694	0.7815	0.842	0.514	1.329
NanoPsu	Psi	883	678	0.7678	0.834	0.513	1.363

Note. Unions over the three independent HeLa replicates of each condition; WT = wild type, unmodified IVT = negative control (not a treatment arm). The 95% CI is the unit-level percentile bootstrap ($B = 1000$, seeds 20260920/20260921) of the mean-of-counts ratio. CHEUI-m5C uses the corrected call set (coordinate fix and reference-base filter, 2026-09-18): the union of 48,627 WT calls quoted in the original submission predates that correction; its IVT/WT union ratio is 1.0717 and its mean-of-counts ratio 1.081. Values for the other tools are unchanged with respect to the published table. The per-unit counts and ratio intervals are deposited in the machine-readable tables that accompany the source code, reconciled row by row against the frozen evidence set behind this table.

Table S6: Per-tool implementation details (software, input, coverage floor, thresholds, multiple-testing correction, coordinate harmonisation).

Tool	Software / version	Required input	Minimum read coverage	Calling threshold	Multiple-testing correction	Default or optimised	Coordinate harmonisation
CHEUI_m6A	Env `cheui`: python 3.7.12, tensorflow-gpu 2.4.1 (tensorflow-base 2.4.1), keras-preprocessing 1.1.2, numpy 1.21.6, pandas 1.3.4...	nanopolish eventalign output (FAST5-derived signal)	CHEUI_predict_model2 default -n/--min_reads = 20 reads per site / Preprocessing...	model probability (default); CHEUI-diff upper_cutoff/lower_cutoff	None for CHEUI-solo (model1/model2 emit probability only, n...	default	transcript -> genomic via BED12 alignme...
DENA	DENA_3.3 / ran in conda env `dena` = python3.7 + ont-tombo 1.5.1 / DENA-public (curlcake runs) no longer on disk	nanopolish eventalign (FAST5-derived signal)	benchmark filter applied post-hoc: coverage (col3+col4) > 20 / tool provides no...	0.1 (manuscript)	none: output has no p-value/q-value column	default	transcript -> genomic
DRUMMER	Env `DRUMMER`: python 3.11.9, scipy 1.14.0, pandas 2.2.2, numpy 1.26.0, biopython 1.83, statsmodels 0.14.2, bedtools 2.26.0 / Ups...	Tombo-resquiggled FAST5	No minimum-depth CLI option; the only depth floor in the pipeline is depth != 0...	FDR-adjusted p-value (GMM-based test)	Benjamini-Hochberg FDR (tool default)	default	transcript -> genomic
ELIGOS2_diff	Same as ELIGOS2_solo: ELIGOS v2.1.0, subcommand pair_diff_mod (container /data/lxy/apptainer/eligos2.sif)	Tombo-resquiggled FAST5	Default --min_depth 20; we used --min_depth 5	FDR-adjusted p-value vs matched IVT/control	FDR (tool default)	default	transcript -> genomic
ELIGOS2_solo	Upstream source https://gitlab.com/piroonj/eligos2 ; container recipe /data/lxy/source_code/benchmark/eligos2-v2.1.0/DockerFiles...	Tombo-resquiggled FAST5	Default --min_depth 20 ("Minimum number of reads") / We used --min_depth 5 (rel...	FDR-adjusted p-value vs background error model	FDR (tool default)	default	transcript -> genomic
EpiNano_Error	1.2.0 vs 1.2.4 DiffErr.R differ by exactly one line / EpiNano 1.2.4 for EVERY EpiNano_DiffErr.R call (0 calls used 1.2.0) / versi...	basecalled FASTQ + BAM (sequence + error features)	tool default -c/--coverage = 30 (strictly greater-than) / upstream extra filter...	per-site error test / SVM score	Bonferroni (via car::outlierTest) on the lm-residual path o...	default	genomic (minimap2 alignments)
m6Anet	Env: python 3.7/3.8 pinned by setup.py, torch==1.6.0 / m6Anet 2.1.0 (__version__ = "2.1.0"); repo has no .git (source release cop...	nanopolish/f5c eventalign + Tombo resquiggle (FAST5-derived)	20 (tool default)	probability >= 0.5 (tool default)	not applicable (probabilistic model)	default; RNA004 uses the RNA004-optimis...	transcript -> genomic
mAFiA	Env mafia runtime: python 3.9.19, torch 1.13.1, scikit-learn 1.2.0, pysam 0.20.0, biopython 1.80, numpy 1.23.5 / The installed pa...	sorted BAM (basecalled)	Our runs: --min_coverage 50 in all 11 invocations (10 in the /data/lxy/benchmark...	model probability (default)	n/a - mAFiA performs no hypothesis test and emits no p-valu...	default	genomic
MINES	MINES: no version string in repo; run as source script cDNA_MINES1.py from /data/lxy/source_code/benchmark/MINES; no .git dir, so...	Tombo-resquiggled FAST5	OUR effective cutoff = coverage >= 20 (post-processing, not the tool) / hard-co...	random-forest probability (default)	none: output has no p-value/q-value column anywhere	default	transcript -> genomic
Nanocompore	1.0.4	SampComp from nanopolish eventalign (FAST5-derived)	5 (auto-extracted from log)	GMM / KS test p-value	BH exists only in the unused SampCompDB path / NONE in the...	default	transcript (GRC m39/gencode transcriptom...
Nanom6A	Nanom6A release "nanom6A_2022_12_23" (vendored copy dir named nanom6A_2022_12_22); no .git dir, no version string in code, `--ver...	sorted BAM + reference (basecalled)	Effective cutoff = int(total_reads * support / 1,000,000) modified reads at the...	0.5 (manuscript)	None - no p-value, q-value or FDR anywhere in the code	default	genomic (minimap2)

Tool	Software / version	Required input	Minimum read coverage	Calling threshold	Multiple-testing correction	Default or optimised	Coordinate harmonisation
NanoMUD	git HEAD (vendored tree, .git_disabled) = d906c5221b09854a70603d5e3b92e80fd65c772d ("Update README.md", 2024-04-15), remote https...	basecalled BAM (sequence-based)	OUR cutoff: coverage > 20 plus Status mod / tool has NO coverage cutoff; it rep...	model probability (default)	none (no p-value; regression/calibration scores only)	default	genomic
NanoNm	NanoNm 1.0.0 (README title) derived from Nanom6A (Gao et al. 2021); vendored copies NanoNm-1.0.0 and NanoNm-main differ only by f...	basecalled BAM + FAST5-derived features	--support default 20 = minimum number of MODIFIED reads per site; we passed --s...	model probability (default)	none (probability + mod-ratio only; ratio.0.5.tsv columns a...	default	genomic
NanoPsu	NanoPsu 1.0 (setup.py version="1.0", bin/nanopsu __version__="1.0"); conda env `nanopsu` / Upstream git commit 01f2ce11d77af22bb4...	basecalled FASTQ/BAM + k-mer features	README: no minimum input-read requirement, but >1M raw reads recommended for a...	model probability (default)	none - no p-value/q-value column exists anywhere in the pac...	default	transcript -> genomic
NanoSPA	NanoSPA 1.0 (setup.py + PKG-INFO + bin/nanospa __version__="1.0"); conda `nanospa=1.0` in env NanoSPA (python 3.10.2, tensorflow...	basecalled FASTQ/BAM + 9-mer features	hard-coded coverage > 20 reads per site, no CLI flag	model probability (default)	none (no p-value column in prediction_m6A.csv / prediction_...	default	transcript -> genomic
Tombo	bioconda build marker / ont-tombo 1.5.1	raw FAST5 (resquiggle)	de_novo default --minimum-test-reads = 1 / level_sample_compare default = 50; w...	statistical test p-value (default)	NOT applied: the BH function exists but is never called	default	transcript (Tombo wiggle)
xPore	setup.py version 'v2.1' vs package __version__ "2.0" / installed from bioconda build dir	f5c/nanopolish eventalign dataprep	20 (auto-extracted from yml)	FDR-adjusted p-value on diff_mod_rate	FDR (tool default)	readcount_min=20, readcount_max=2000000	transcript -> genomic
Yanocomp	0.2	Tombo-resquiggled FAST5	benchmark value: -n 50 (all datasets except the curlcake runs, which left it dy...	KS/GMM test p-value	Benjamini-Hochberg (statsmodels) applied inside gmmtest / f...	default	transcript -> genomic
differr	0.2 (setup.py), editable install of the vendored tree	basecalled BAM + reference	--median-expr-threshold default 10 / --min-expr-threshold default 1 (all replic...	error-rate test p-value	Benjamini-Hochberg (statsmodels) on the pooled-comparison G...	default	genomic
Dorado (RNA004 modification models)	0.9.1 (banner from /data/lxy/tool/dorado-0.9.1-linux-x64/bin/dorado --version)	POD5 -> basecalling with --modified-bases-models	10 (applied in this study)	percent_modified >= 5/10/20/50 (threshold scan)	not applicable	default model checkpoints	genomic (modkit pileup)

Note. Generated from the artifacts of this benchmark and from the tool documentation; every curated cell is marked in the evidence column of the machine-readable table deposited with the source code, which carries the full text without clipping. The RNA004 Dorado models and the RNA004-optimised m6Anet model are included.

Table S7: Applicability of each tool to the RNA002 and RNA004 chemistries, with the failure or exclusion class where a tool was not applied or not included.

Tool	Input class	Used on RNA002	Used on RNA004	Status	Failure or exclusion class
CHEUI_m6A	raw-signal (nanopolish eventalign)	yes	no	not applied to RNA004	input format: raw signal (a nanopolish/f5c eventalign or Tombo resquiggle) cannot be produced from RNA004 data
CHEUI-diff	not determined	no	no	not included in this benchmark	outside the scope of this study: output generated but not used
DENA	raw-signal (nanopolish eventalign)	yes	no	not applied to RNA004	input format: raw signal (a nanopolish/f5c eventalign or Tombo resquiggle) cannot be produced from RNA004 data
differr	basecalled (BAM)	no	no	not included in this benchmark	outside the scope of this study: output generated but not used
DRUMMER	raw-signal (Tombo resquiggle)	yes	yes	included in the RNA004 benchmark	
ELIGOS2_diff	raw-signal (Tombo resquiggle)	yes	yes	included in the RNA004 benchmark	
ELIGOS2_solo	raw-signal (Tombo resquiggle)	yes	yes	included in the RNA004 benchmark	
EpiNano_Error	basecalled (BAM/FASTQ)	yes	no	not applied to RNA004	basecalling model: released model trained on RNA002 features; no RNA004 checkpoint evaluated in this study
Epinano_SVM	basecalled (BAM/FASTQ)	no	no	not included in this benchmark	outside the scope of this study: output generated but not used
m6Anet	raw-signal (eventalign) / RNA004 BAM model	yes	yes	included in the RNA004 benchmark	
mAFIA	basecalled (BAM)	no	no	not included in this benchmark	outside the scope of this study: output generated but not used
MINES	raw-signal (Tombo resquiggle)	yes	no	not applied to RNA004	basecalling model: released model trained on RNA002 features; no RNA004 checkpoint evaluated in this study
Nanocompore	raw-signal (eventalign)	yes	no	not applied to RNA004	input format: raw signal (a nanopolish/f5c eventalign or Tombo resquiggle) cannot be produced from RNA004 data
Nanom6A	basecalled (BAM)	yes	no	not applied to RNA004	basecalling model: released model trained on RNA002 features; no RNA004 checkpoint evaluated in this study
NanoMUD	basecalled (BAM)	yes	no	not applied to RNA004	basecalling model: released model trained on RNA002 features; no RNA004 checkpoint evaluated in this study
NanoNm	basecalled (BAM)	yes	no	not applied to RNA004	basecalling model: released model trained on RNA002 features; no RNA004 checkpoint evaluated in this study
NanoPsu	basecalled (FASTQ/BAM)	yes	yes	included in the RNA004 benchmark	
NanoSPA	basecalled (FASTQ/BAM)	yes	yes	included in the RNA004 benchmark	
Tombo	raw-signal (FAST5)	no	no	not applied to RNA004	software maintenance: the resquiggle component is not maintained for the new chemistry
Tombo (comparative)	not determined	no	no	not included in this benchmark	outside the scope of this study: output generated but not used
xPore	raw-signal (eventalign)	yes	no	not applied to RNA004	input format: raw signal (a nanopolish/f5c eventalign or Tombo resquiggle) cannot be produced from RNA004 data

Tool	Input class	Used on RNA002	Used on RNA004	Status	Failure or exclusion class
Yanocomp	raw-signal (Tombo resquiggle)	yes	no	not applied to RNA004	basecalling model: released model trained on RNA002 features; no RNA004 checkpoint evaluated in this study

Note. Presence on a chemistry is read from the file system (call directories per tool and sample), not from memory. Input classes: raw-signal (FAST5 plus a nanopolish/Tombo resquiggle) and basecalled (BAM/FASTQ) callers; RNA004 native output is POD5 and Dorado BAM with 9-mer models, so RNA002 signal models cannot be produced for it. Failure and exclusion classes: input format (raw signal not producible from RNA004 data), basecalling model (RNA002-trained model, no RNA004 checkpoint evaluated), software maintenance (resquiggle not maintained for the new chemistry), and outside the scope of this study (output generated but not used).

Table S8: Biological source of each GLORI reference relative to the matched nanopore dataset.

Species	Nanopore libraries	GLORI reference	Biological relationship	GLORI sites
Human	HeLa WT (SRP393373, RNA002) and HeLa WT/IVT (ERP164259, RNA004)	HeLa, GSE210563 (Liu et al. 2023)	Same cell line, independent laboratory and independent RNA extract	112,451
Arabidopsis	Col-0 WT (SRP329449) and fip37 KD (SRP363914), RNA002	Col-0 seedlings, GSE246632 (Xie et al. 2025)	Same species and accession from independent studies and independent RNA extracts	80,624
Mouse	mESC Mettl3 KO (SRP166020) and mESC WT (SRP357195), RNA002	mESC, GSE246632	Same cell type (mESC) from independent studies and independent RNA extracts	41,961

Note. Counts are the replicate-overlapping GLORI sites with modification ratio > 0.1 in both replicates (Fig. S3A). GSE210563 also contains HEK293T GLORI, and its mouse samples are MEF; neither was used. Nanopore and GLORI libraries were never derived from the same RNA extract, so a GLORI-positive site is not assumed to be modified in the corresponding nanopore sample and the PPV against GLORI is a precision-like measure conditioned on this site-level partial reference, not sensitivity.

Table S9: Positive and negative references available for each modification type, and the metrics that can therefore be reported.

Modification	Positive reference	Negative control	Metrics reported	Limitation stated in the manuscript
m6A	GLORI (site-level partial reference; Table S8)	Unmodified Curlcake (SRP174366, ERP162788) and HeLa IVT	PPV against GLORI, recall, F1, MCC, false-positive burden, replicate consistency	GLORI is an independently generated, DRACH-centred and coverage-limited reference; absolute values are reference-bounded.
Psi	None (no orthogonal Psi map is used for scoring)	Unmodified Curlcake and HeLa IVT	False-positive burden, replicate consistency, enrichment over a chromosome-stratified permutation null against RMBase + DirectRMDb	Sensitivity and recall are not reported; the m6A-centred GLORI map carries no Psi labels.
m1Psi	None	Unmodified Curlcake and HeLa IVT	False-positive burden, replicate consistency	No external reference covers m1Psi (RMBase + DirectRMDb covers Psi, m5C and Nm only).
m5C	None (orthogonal UBS-seq m5C used only as an external overlap set)	Unmodified Curlcake and HeLa IVT	False-positive burden, replicate consistency, enrichment over the permutation null	The synthetic constructs are unmodified or fully m6A-substituted, so no m5C-positive control is available.
Nm	None (orthogonal Nm-Mut-seq used only as an external overlap set)	Unmodified Curlcake and HeLa IVT	False-positive burden, replicate consistency, enrichment over the permutation null	Same as m5C: no synthetic Nm-positive control, and GLORI provides no Nm labels.

Note. The Curlcake libraries used here are the unmodified IVT set and the fully m6A-substituted set (SRP174366 / GSE124309) plus the unmodified RNA004 set (ERP162788); later Curlcake collections that spike in Psi, m5C or m1Psi were not among the deposited accessions and were not used.

Table S10: Datasets used in this benchmark and their replicate structure.

Species	Dataset	Chemistry	Study	Samples	n	Role
Human	HeLa, WT	RNA002	SRP393373	HeLa_WT1, HeLa_WT2, HeLa_WT3	3	WT reference
Human	HeLa, IVT	RNA002	SRP428418	HeLa_IVT_rep1, HeLa_IVT_rep2, HeLa_IVT_rep3	3	negative control
Arabidopsis	WT (Col-0)	RNA002	SRP329449	Arabidopsis_WT_rep1, Arabidopsis_WT_rep2, Arabidopsis_WT_rep3	3	WT reference
Arabidopsis	FIP37 KD	RNA002	SRP363914	Arabidopsis_fip37_rep1, Arabidopsis_fip37_rep2, Arabidopsis_fip37_rep3	3	partial negative
Mouse	WT (mESC)	RNA002	SRP166020	mESCs_Mettl3_WT	1	WT reference
Mouse	WT (mESC)	RNA002	SRP357195	mES_WT	1	WT reference
Mouse	Mettl3 KO (mESC)	RNA002	SRP166020	mESCs_Mettl3_KO	1	partial negative
Mouse	Mettl3 KO (mESC)	RNA002	SRP357195	mES_KO	1	partial negative
E.coli	WT (K-12)	RNA002	SRP478171	E_ss_rd_RNA1, E_ss_rd_RNA2	2	WT reference (null vs null)
E.coli	IVT	RNA002	SRP478171	E_IVT_neg1, E_IVT_neg2	1	negative control (null vs null)
Curlicake	IVT (unmodified)	RNA002	SRP174366	Curlicake_IVT_rep1, Curlicake_IVT_rep2_partial, Curlicake_IVT_rep3	2	negative control
Curlicake	fully m6A-modified	RNA002	SRP174366	Curlicake_m6A_rep1, Curlicake_m6A_rep2	2	positive control
Human	HeLa, WT	RNA004	ERP164259	HeLa_RNA004_WT	1	WT reference
Human	HeLa, IVT	RNA004	ERP164259	HeLa_RNA004_IVT	1	negative control
Curlicake	IVT (unmodified)	RNA004	ERP162788	Curlicake_RNA004_IVT	1	negative control

Note. Each row is one dataset (species x condition x chemistry). The two mouse studies are independent studies, not replicates, and are never pooled; the last column gives the number of independent sequencing units used in the site-level evaluation (biological replicates for HeLa, Arabidopsis and Curlicake; single units for mouse, E. coli IVT and the RNA004 datasets). Curlicake_IVT_rep2_partial is a depth-matched subset of the full run and is not counted as a replicate; the machine-readable sample registry deposited with the source code carries the per-sample provenance and accession of every row.

Table S11: Tool nomenclature used throughout the manuscript, figures and Supporting Information.

Standardized name	Tool	Configuration evaluated	Software version / model
xPore	xPore	comparative (WT vs control)	setup.py version 'v2.1' vs package __version__ "2.0"
Nanom6A	Nanom6A	de novo	Nanom6A release "nanom6A_2022_12_23" (vendored copy dir named nanom6A_2
ELIGOS2_solo	ELIGOS2	differential (ELIGOS2_diff) and single-sample (ELIGOS2_solo)	not recorded
ELIGOS2_diff	ELIGOS2	differential (ELIGOS2_diff) and single-sample (ELIGOS2_solo)	not recorded
DRUMMER	DRUMMER	comparative (WT vs control)	Env `DRUMMER`: python 3.11.9, scipy 1.14.0, pandas 2.2.2, numpy 1.26.0,
Nanocompore	Nanocompore	comparative (WT vs control)	1.0.4
EpiNano_Error	EpiNano_Error		1.2.0 vs 1.2.4 DiffErr.R differ by exactly one line EpiNan
m6Anet	m6Anet	de novo (RNA002 and RNA004-optimised models)	Env: python 3.7/3.8 pinned by setup.py, torch==1.6.0
MINES	MINES	de novo	MINES: no version string in repo
DENA	DENA	de novo	DENA_3.3
Yanocomp	Yanocomp	de novo	0.2
CHEUI_m6A	CHEUI	CHEUI_m6A, CHEUI_m5C	not recorded
CHEUI_m5C	CHEUI	CHEUI_m6A, CHEUI_m5C	not recorded
NanoSPA_m6A	NanoSPA	NanoSPA_m6A, NanoSPA_Psi	NanoSPA 1.0 (setup.py + PKG-INFO + bin/nanospa __version__="1.0")
NanoSPA_Psi	NanoSPA	NanoSPA_m6A, NanoSPA_Psi	NanoSPA 1.0 (setup.py + PKG-INFO + bin/nanospa __version__="1.0")
NanoMUD_Psi	NanoMUD	NanoMUD_Psi, NanoMUD_m1Psi	git HEAD (vendored tree, .git_disabled) = d906c5221b09854a70603d5e3b92e
NanoMUD_m1Psi	NanoMUD	NanoMUD_Psi, NanoMUD_m1Psi	git HEAD (vendored tree, .git_disabled) = d906c5221b09854a70603d5e3b92e
NanoPsu	NanoPsu	de novo	NanoPsu 1.0 (setup.py version="1.0", bin/nanopsu __version__="1.0")
NanoNm	NanoNm	de novo	NanoNm 1.0.0 (README title) derived from Nanom6A (Gao et al. 2021)

Note. names are standardized as in the manuscript; the reference for every tool is given in the numbered reference list of the manuscript. Technical parameters and the coordinate harmonisation are in Table S6.

Table S12: Sensitivity of the site-level tool ranking to transcript coverage and to the reference's own composition.

Species	Units	Sensitivity axis	Level	n tools	Spearman rho vs primary	BH-FDR	Top-ranked tool identical	Top-ranked tool
Arabidopsis	3	coverage floor	coverage floor >= 5 reads	13	0.999	5.5e-15	yes	m6Anet
Arabidopsis	3	coverage floor	coverage floor >= 20 reads	13	1.000	< 1e-16	yes	m6Anet
Arabidopsis	3	coverage floor	coverage floor >= 50 reads	13	0.876	0.00011	yes	m6Anet
Arabidopsis	3	coverage floor	coverage floor >= 100 reads	13	0.629	0.022	yes	m6Anet
Arabidopsis	3	coverage stratum	stratum 5-9 reads	10	0.835	0.0048	no	DENA
Arabidopsis	3	coverage stratum	stratum 10-19 reads	11	0.879	0.00083	no	DENA
Arabidopsis	3	coverage stratum	stratum 20-49 reads	13	0.801	0.002	yes	m6Anet
Arabidopsis	3	coverage stratum	stratum >= 50 reads	13	0.876	0.00025	yes	m6Anet
Arabidopsis	3	reference ratio stratum	reference ratio 0.1-0.3	13	0.581	0.039	no	MINES
Arabidopsis	3	reference ratio stratum	reference ratio 0.3-0.6	13	0.737	0.0048	no	ELIGOS2_solo
Arabidopsis	3	reference ratio stratum	reference ratio > 0.6	13	0.845	0.00044	no	ELIGOS2_solo
Mouse	1	coverage floor	coverage floor >= 5 reads	13	1.000	< 1e-16	yes	m6Anet
Mouse	1	coverage floor	coverage floor >= 20 reads	13	0.995	9.4e-12	yes	m6Anet
Mouse	1	coverage floor	coverage floor >= 50 reads	13	0.918	1.4e-05	yes	m6Anet
Mouse	1	coverage floor	coverage floor >= 100 reads	13	0.830	0.00051	no	ELIGOS2_diff
Mouse	1	coverage stratum	stratum 5-9 reads	10	0.267	0.56	no	NanoSPA_m6A
Mouse	1	coverage stratum	stratum 10-19 reads	12	0.699	0.018	no	DENA
Mouse	1	coverage stratum	stratum 20-49 reads	13	0.874	0.00025	yes	m6Anet
Mouse	1	coverage stratum	stratum >= 50 reads	13	0.918	4e-05	yes	m6Anet
Mouse	1	reference ratio stratum	reference ratio 0.1-0.3	13	0.478	0.1	yes	m6Anet
Mouse	1	reference ratio stratum	reference ratio 0.3-0.6	13	0.714	0.0068	yes	m6Anet
Mouse	1	reference ratio stratum	reference ratio > 0.6	13	0.720	0.0064	yes	m6Anet
Human	3	coverage floor	coverage floor >= 5 reads	13	1.000	< 1e-16	yes	m6Anet
Human	3	coverage floor	coverage floor >= 20 reads	13	1.000	< 1e-16	yes	m6Anet
Human	3	coverage floor	coverage floor >= 50 reads	13	0.985	2.2e-09	yes	m6Anet
Human	3	coverage floor	coverage floor >= 100 reads	13	0.884	7.8e-05	yes	m6Anet
Human	3	coverage stratum	stratum 5-9 reads	9	0.119	0.79	no	ELIGOS2_diff
Human	3	coverage stratum	stratum 10-19 reads	11	0.442	0.24	no	ELIGOS2_diff
Human	3	coverage stratum	stratum 20-49 reads	13	0.968	5.5e-07	no	Nanom6A
Human	3	coverage stratum	stratum >= 50 reads	13	0.985	1.7e-08	yes	m6Anet
Human	3	reference ratio stratum	reference ratio 0.1-0.3	13	0.743	0.0045	no	MINES
Human	3	reference ratio stratum	reference ratio 0.3-0.6	13	0.899	5.2e-05	no	Nanom6A
Human	3	reference ratio stratum	reference ratio > 0.6	13	0.904	4.3e-05	no	Nanom6A

Note. Ranking statistic MCC (the metric that orders the rows of Fig. 5A), computed per independent sequencing unit inside each sample's own candidate-site set and averaged across units, exactly as in the main analysis; each row restricts the evaluation to the listed level and compares the resulting per-unit mean rank with the primary ranking (13 m6A tool configurations, candidate-site set with coverage >= 10 reads, full high-confidence reference). Spearman rho is that rank

correlation and BH-FDR its correction within the axis. Arabidopsis and HeLa contribute three units, the mouse study one, so no interval is reported for mouse. The three coverage floors that leave the evaluation set comparable (≥ 5 , ≥ 10 and ≥ 20 reads) give the same ordering ($\rho = 0.99-1.00$, top-ranked tool unchanged in every species); inside the informative coverage strata (20-49 and ≥ 50 reads) the ranking is reproduced at $\rho = 0.80-0.98$. The sparse strata (5-9 and 10-19 reads) carry almost no information - the mean recall of every tool is $\leq 0.9\%$ there - and are reported as sensitivity checks only. Stratifying the reference by modification ratio changes the ordering more strongly ($\rho = 0.48-0.90$, with the top-ranked tool changing in seven of the nine rows), which is why the primary metrics are reported against the full high-confidence reference.